# Table 15 Reduced Form Effects on WHO Anthropometric Outcomes for Ages 0-6 Using 10 October 2022 Child Age

The child cohort uses complete birth dates and completed ages 0 to 83 months on 10 October 2022 without age-boundary assignments. WHO 2006 covers months 0-60 and WHO 2007 covers months 61 onward. Weight-for-length/height classifications are available through month 60 only. Father RDD estimates are the primary analysis and mother RDD estimates are secondary.

### Father RDD primary analysis

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.068 | **0.294* [0.163] N=2,728** | **0.334* [0.171] N=2,517** | **0.384** [0.186] N=2,249** | **0.501** [0.192] N=1,936** | **0.476** [0.219] N=1,598** |
| WHO weight-for-age z score | 0.307 | -0.052 [0.094] N=3,017 | -0.034 [0.103] N=2,784 | 0.033 [0.102] N=2,490 | -0.006 [0.112] N=2,138 | -0.000 [0.120] N=1,762 |
| WHO weight-for-length/height z score, 0-60 months | 0.351 | 0.038 [0.155] N=2,055 | 0.031 [0.158] N=1,908 | 0.039 [0.168] N=1,720 | -0.153 [0.185] N=1,474 | -0.082 [0.197] N=1,208 |
| WHO BMI-for-age z score | 0.353 | 0.011 [0.143] N=2,642 | 0.007 [0.146] N=2,441 | 0.024 [0.157] N=2,177 | -0.231 [0.163] N=1,871 | -0.211 [0.175] N=1,544 |
| Stunted: HAZ below -2 | 0.187 P=494 | -0.035 [0.032] N=2,728 | -0.039 [0.033] N=2,517 | -0.041 [0.034] N=2,249 | **-0.060* [0.035] N=1,936** | -0.047 [0.041] N=1,598 |
| Underweight: WAZ below -2 | 0.054 P=164 | 0.014 [0.017] N=3,017 | 0.008 [0.018] N=2,784 | -0.001 [0.019] N=2,490 | -0.010 [0.021] N=2,138 | -0.018 [0.025] N=1,762 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=230 | -0.025 [0.026] N=2,055 | -0.021 [0.027] N=1,908 | -0.025 [0.029] N=1,720 | -0.006 [0.031] N=1,474 | -0.027 [0.032] N=1,208 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.199 P=403 | 0.010 [0.036] N=2,055 | 0.006 [0.038] N=1,908 | 0.010 [0.040] N=1,720 | -0.016 [0.045] N=1,474 | -0.022 [0.052] N=1,208 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.092 P=191 | -0.008 [0.028] N=2,055 | -0.011 [0.030] N=1,908 | 0.004 [0.032] N=1,720 | -0.013 [0.036] N=1,474 | -0.014 [0.042] N=1,208 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=513 | -0.044 [0.037] N=1,987 | -0.037 [0.039] N=1,845 | -0.032 [0.040] N=1,662 | -0.053 [0.042] N=1,424 | -0.037 [0.046] N=1,167 |
| BMI-for-age z score below -2 | 0.115 P=307 | 0.000 [0.027] N=2,642 | -0.001 [0.027] N=2,441 | -0.002 [0.030] N=2,177 | 0.029 [0.032] N=1,871 | 0.029 [0.035] N=1,544 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.369 P=221 | -0.021 [0.075] N=586 | -0.035 [0.078] N=531 | -0.039 [0.084] N=461 | -0.053 [0.088] N=405 | -0.106 [0.102] N=339 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.222 P=132 | -0.012 [0.061] N=586 | -0.038 [0.062] N=531 | -0.025 [0.071] N=461 | -0.053 [0.072] N=405 | -0.122 [0.078] N=339 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=62 | 0.046 [0.072] N=586 | 0.054 [0.076] N=531 | 0.033 [0.079] N=461 | 0.026 [0.088] N=405 | 0.017 [0.111] N=339 |

*Specification 1 Monthly child-age fixed effects 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.021 [0.215] N=2,289 | -0.013 [0.236] N=2,078 | 0.016 [0.270] N=1,810 | 0.218 [0.288] N=1,497 | -0.061 [0.382] N=1,159 |
| WHO weight-for-age z score | **-0.218* [0.130] N=2,539** | -0.212 [0.150] N=2,306 | -0.108 [0.154] N=2,012 | -0.226 [0.186] N=1,660 | -0.292 [0.233] N=1,284 |
| WHO weight-for-length/height z score, 0-60 months | 0.052 [0.222] N=1,731 | 0.051 [0.235] N=1,584 | 0.027 [0.266] N=1,396 | -0.369 [0.325] N=1,150 | -0.326 [0.410] N=884 |
| WHO BMI-for-age z score | 0.181 [0.211] N=2,216 | 0.218 [0.221] N=2,015 | 0.293 [0.260] N=1,751 | -0.142 [0.299] N=1,445 | -0.079 [0.389] N=1,118 |
| Stunted: HAZ below -2 | 0.032 [0.043] N=2,289 | 0.041 [0.046] N=2,078 | 0.054 [0.049] N=1,810 | 0.037 [0.058] N=1,497 | **0.144* [0.077] N=1,159** |
| Underweight: WAZ below -2 | **0.039* [0.022] N=2,539** | 0.034 [0.025] N=2,306 | 0.022 [0.027] N=2,012 | 0.011 [0.033] N=1,660 | 0.005 [0.045] N=1,284 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.023 [0.036] N=1,731 | -0.017 [0.039] N=1,584 | -0.020 [0.044] N=1,396 | 0.013 [0.055] N=1,150 | -0.035 [0.063] N=884 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.010 [0.044] N=1,731 | 0.005 [0.047] N=1,584 | 0.006 [0.052] N=1,396 | -0.053 [0.065] N=1,150 | -0.105 [0.088] N=884 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.018 [0.030] N=1,731 | -0.022 [0.034] N=1,584 | 0.004 [0.038] N=1,396 | -0.036 [0.045] N=1,150 | -0.066 [0.064] N=884 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.012 [0.050] N=1,672 | 0.033 [0.055] N=1,530 | 0.055 [0.057] N=1,347 | 0.021 [0.068] N=1,109 | 0.115 [0.082] N=852 |
| BMI-for-age z score below -2 | -0.027 [0.032] N=2,216 | -0.037 [0.032] N=2,015 | -0.050 [0.038] N=1,751 | -0.008 [0.043] N=1,445 | -0.019 [0.052] N=1,118 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.058 [0.120] N=481 | 0.059 [0.129] N=426 | 0.099 [0.138] N=356 | 0.133 [0.161] N=300 | 0.044 [0.215] N=234 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.094 [0.091] N=481 | 0.074 [0.098] N=426 | 0.140 [0.115] N=356 | 0.138 [0.130] N=300 | 0.007 [0.145] N=234 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.014 [0.066] N=481 | 0.020 [0.070] N=426 | -0.035 [0.070] N=356 | -0.071 [0.087] N=300 | -0.155 [0.130] N=234 |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.068 | **0.297* [0.163] N=2,728** | **0.336* [0.171] N=2,517** | **0.389** [0.186] N=2,249** | **0.506*** [0.190] N=1,936** | **0.481** [0.215] N=1,598** |
| WHO weight-for-age z score | 0.307 | -0.052 [0.094] N=3,017 | -0.034 [0.103] N=2,784 | 0.033 [0.102] N=2,490 | -0.004 [0.112] N=2,138 | -0.000 [0.120] N=1,762 |
| WHO weight-for-length/height z score, 0-60 months | 0.351 | 0.038 [0.155] N=2,055 | 0.031 [0.159] N=1,908 | 0.037 [0.169] N=1,720 | -0.153 [0.186] N=1,474 | -0.090 [0.198] N=1,208 |
| WHO BMI-for-age z score | 0.353 | 0.012 [0.143] N=2,642 | 0.007 [0.146] N=2,441 | 0.025 [0.158] N=2,177 | -0.229 [0.164] N=1,871 | -0.211 [0.175] N=1,544 |
| Stunted: HAZ below -2 | 0.187 P=494 | -0.035 [0.032] N=2,728 | -0.039 [0.033] N=2,517 | -0.040 [0.034] N=2,249 | **-0.060* [0.035] N=1,936** | -0.047 [0.041] N=1,598 |
| Underweight: WAZ below -2 | 0.054 P=164 | 0.014 [0.017] N=3,017 | 0.008 [0.018] N=2,784 | -0.001 [0.019] N=2,490 | -0.010 [0.021] N=2,138 | -0.018 [0.025] N=1,762 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=230 | -0.023 [0.026] N=2,055 | -0.018 [0.027] N=1,908 | -0.021 [0.029] N=1,720 | -0.002 [0.031] N=1,474 | -0.021 [0.032] N=1,208 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.199 P=403 | 0.011 [0.036] N=2,055 | 0.008 [0.038] N=1,908 | 0.013 [0.040] N=1,720 | -0.012 [0.045] N=1,474 | -0.018 [0.051] N=1,208 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.092 P=191 | -0.007 [0.028] N=2,055 | -0.008 [0.030] N=1,908 | 0.007 [0.032] N=1,720 | -0.010 [0.035] N=1,474 | -0.010 [0.041] N=1,208 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=513 | -0.041 [0.037] N=1,987 | -0.033 [0.038] N=1,845 | -0.027 [0.039] N=1,662 | -0.048 [0.042] N=1,424 | -0.030 [0.045] N=1,167 |
| BMI-for-age z score below -2 | 0.115 P=307 | 0.001 [0.026] N=2,642 | 0.001 [0.027] N=2,441 | -0.001 [0.030] N=2,177 | 0.030 [0.032] N=1,871 | 0.030 [0.035] N=1,544 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.369 P=221 | -0.022 [0.075] N=586 | -0.036 [0.078] N=531 | -0.042 [0.084] N=461 | -0.061 [0.088] N=405 | -0.111 [0.103] N=339 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.222 P=132 | -0.014 [0.061] N=586 | -0.040 [0.063] N=531 | -0.029 [0.073] N=461 | -0.064 [0.074] N=405 | **-0.133* [0.079] N=339** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=62 | 0.047 [0.073] N=586 | 0.055 [0.076] N=531 | 0.034 [0.079] N=461 | 0.027 [0.086] N=405 | 0.018 [0.108] N=339 |

*Specification 2 Monthly fixed effects and female 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.022 [0.214] N=2,289 | -0.013 [0.235] N=2,078 | 0.015 [0.267] N=1,810 | 0.200 [0.285] N=1,497 | -0.117 [0.375] N=1,159 |
| WHO weight-for-age z score | **-0.219* [0.130] N=2,539** | -0.211 [0.150] N=2,306 | -0.108 [0.154] N=2,012 | -0.230 [0.186] N=1,660 | -0.299 [0.231] N=1,284 |
| WHO weight-for-length/height z score, 0-60 months | 0.052 [0.222] N=1,731 | 0.050 [0.236] N=1,584 | 0.025 [0.266] N=1,396 | -0.370 [0.326] N=1,150 | -0.330 [0.411] N=884 |
| WHO BMI-for-age z score | 0.181 [0.211] N=2,216 | 0.218 [0.221] N=2,015 | 0.292 [0.260] N=1,751 | -0.144 [0.300] N=1,445 | -0.070 [0.390] N=1,118 |
| Stunted: HAZ below -2 | 0.033 [0.043] N=2,289 | 0.041 [0.046] N=2,078 | 0.054 [0.049] N=1,810 | 0.039 [0.058] N=1,497 | **0.151* [0.076] N=1,159** |
| Underweight: WAZ below -2 | **0.039* [0.023] N=2,539** | 0.034 [0.025] N=2,306 | 0.022 [0.027] N=2,012 | 0.011 [0.033] N=1,660 | 0.005 [0.045] N=1,284 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.023 [0.037] N=1,731 | -0.016 [0.039] N=1,584 | -0.018 [0.044] N=1,396 | 0.014 [0.055] N=1,150 | -0.034 [0.064] N=884 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.010 [0.044] N=1,731 | 0.006 [0.047] N=1,584 | 0.007 [0.052] N=1,396 | -0.052 [0.064] N=1,150 | -0.104 [0.087] N=884 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.018 [0.030] N=1,731 | -0.022 [0.034] N=1,584 | 0.005 [0.038] N=1,396 | -0.035 [0.045] N=1,150 | -0.065 [0.064] N=884 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.012 [0.049] N=1,672 | 0.033 [0.054] N=1,530 | 0.055 [0.057] N=1,347 | 0.020 [0.068] N=1,109 | 0.114 [0.082] N=852 |
| BMI-for-age z score below -2 | -0.027 [0.032] N=2,216 | -0.037 [0.032] N=2,015 | -0.049 [0.038] N=1,751 | -0.008 [0.043] N=1,445 | -0.022 [0.052] N=1,118 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.059 [0.120] N=481 | 0.059 [0.130] N=426 | 0.096 [0.140] N=356 | 0.122 [0.163] N=300 | 0.063 [0.219] N=234 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.095 [0.092] N=481 | 0.074 [0.098] N=426 | 0.134 [0.115] N=356 | 0.115 [0.129] N=300 | -0.021 [0.141] N=234 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.014 [0.066] N=481 | 0.020 [0.069] N=426 | -0.031 [0.070] N=356 | -0.057 [0.086] N=300 | -0.123 [0.126] N=234 |

*Specification 3 Monthly fixed effects female and household income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.068 | **0.277* [0.166] N=2,728** | **0.316* [0.174] N=2,517** | **0.362* [0.191] N=2,249** | **0.484** [0.195] N=1,936** | **0.434* [0.221] N=1,598** |
| WHO weight-for-age z score | 0.307 | -0.084 [0.092] N=3,017 | -0.062 [0.102] N=2,784 | 0.000 [0.101] N=2,490 | -0.030 [0.111] N=2,138 | -0.032 [0.120] N=1,762 |
| WHO weight-for-length/height z score, 0-60 months | 0.351 | 0.033 [0.158] N=2,055 | 0.031 [0.162] N=1,908 | 0.043 [0.171] N=1,720 | -0.146 [0.188] N=1,474 | -0.087 [0.201] N=1,208 |
| WHO BMI-for-age z score | 0.353 | 0.005 [0.144] N=2,642 | 0.003 [0.147] N=2,441 | 0.024 [0.158] N=2,177 | -0.227 [0.164] N=1,871 | -0.210 [0.177] N=1,544 |
| Stunted: HAZ below -2 | 0.187 P=494 | -0.032 [0.032] N=2,728 | -0.035 [0.033] N=2,517 | -0.035 [0.035] N=2,249 | -0.056 [0.036] N=1,936 | -0.040 [0.042] N=1,598 |
| Underweight: WAZ below -2 | 0.054 P=164 | 0.018 [0.017] N=3,017 | 0.013 [0.018] N=2,784 | 0.005 [0.018] N=2,490 | -0.005 [0.021] N=2,138 | -0.010 [0.025] N=1,762 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=230 | -0.019 [0.026] N=2,055 | -0.013 [0.027] N=1,908 | -0.017 [0.029] N=1,720 | 0.001 [0.031] N=1,474 | -0.017 [0.032] N=1,208 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.199 P=403 | 0.015 [0.036] N=2,055 | 0.013 [0.038] N=1,908 | 0.017 [0.040] N=1,720 | -0.008 [0.045] N=1,474 | -0.012 [0.052] N=1,208 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.092 P=191 | -0.005 [0.028] N=2,055 | -0.006 [0.030] N=1,908 | 0.009 [0.032] N=1,720 | -0.008 [0.035] N=1,474 | -0.009 [0.041] N=1,208 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=513 | -0.031 [0.037] N=1,987 | -0.020 [0.038] N=1,845 | -0.018 [0.039] N=1,662 | -0.041 [0.042] N=1,424 | -0.019 [0.046] N=1,167 |
| BMI-for-age z score below -2 | 0.115 P=307 | 0.003 [0.026] N=2,642 | 0.003 [0.027] N=2,441 | 0.002 [0.030] N=2,177 | 0.032 [0.031] N=1,871 | 0.033 [0.035] N=1,544 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.369 P=221 | -0.022 [0.076] N=586 | -0.034 [0.079] N=531 | -0.041 [0.086] N=461 | -0.059 [0.091] N=405 | -0.111 [0.107] N=339 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.222 P=132 | -0.018 [0.062] N=586 | -0.044 [0.065] N=531 | -0.034 [0.074] N=461 | -0.068 [0.076] N=405 | -0.133 [0.082] N=339 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=62 | 0.047 [0.073] N=586 | 0.052 [0.077] N=531 | 0.036 [0.079] N=461 | 0.029 [0.087] N=405 | 0.021 [0.109] N=339 |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.035 [0.216] N=2,289 | -0.032 [0.234] N=2,078 | -0.014 [0.269] N=1,810 | 0.174 [0.287] N=1,497 | -0.187 [0.369] N=1,159 |
| WHO weight-for-age z score | **-0.248* [0.127] N=2,539** | -0.235 [0.147] N=2,306 | -0.138 [0.150] N=2,012 | -0.254 [0.182] N=1,660 | -0.337 [0.228] N=1,284 |
| WHO weight-for-length/height z score, 0-60 months | 0.046 [0.225] N=1,731 | 0.052 [0.241] N=1,584 | 0.035 [0.268] N=1,396 | -0.371 [0.328] N=1,150 | -0.349 [0.415] N=884 |
| WHO BMI-for-age z score | 0.173 [0.212] N=2,216 | 0.213 [0.222] N=2,015 | 0.296 [0.260] N=1,751 | -0.141 [0.299] N=1,445 | -0.076 [0.392] N=1,118 |
| Stunted: HAZ below -2 | 0.035 [0.043] N=2,289 | 0.044 [0.046] N=2,078 | 0.058 [0.049] N=1,810 | 0.040 [0.057] N=1,497 | **0.159** [0.076] N=1,159** |
| Underweight: WAZ below -2 | **0.043* [0.023] N=2,539** | 0.038 [0.026] N=2,306 | 0.026 [0.027] N=2,012 | 0.014 [0.034] N=1,660 | 0.013 [0.045] N=1,284 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.017 [0.037] N=1,731 | -0.010 [0.040] N=1,584 | -0.015 [0.045] N=1,396 | 0.018 [0.056] N=1,150 | -0.026 [0.067] N=884 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.014 [0.045] N=1,731 | 0.012 [0.048] N=1,584 | 0.011 [0.053] N=1,396 | -0.051 [0.065] N=1,150 | -0.095 [0.087] N=884 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.017 [0.030] N=1,731 | -0.019 [0.035] N=1,584 | 0.006 [0.038] N=1,396 | -0.036 [0.045] N=1,150 | -0.068 [0.063] N=884 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.023 [0.049] N=1,672 | 0.048 [0.054] N=1,530 | 0.062 [0.056] N=1,347 | 0.020 [0.067] N=1,109 | 0.122 [0.082] N=852 |
| BMI-for-age z score below -2 | -0.024 [0.032] N=2,216 | -0.035 [0.033] N=2,015 | -0.048 [0.039] N=1,751 | -0.007 [0.043] N=1,445 | -0.015 [0.053] N=1,118 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.058 [0.119] N=481 | 0.067 [0.131] N=426 | 0.095 [0.142] N=356 | 0.115 [0.165] N=300 | 0.053 [0.220] N=234 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.091 [0.091] N=481 | 0.074 [0.098] N=426 | 0.121 [0.114] N=356 | 0.100 [0.127] N=300 | -0.030 [0.143] N=234 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.016 [0.065] N=481 | 0.017 [0.070] N=426 | -0.027 [0.070] N=356 | -0.043 [0.084] N=300 | -0.120 [0.129] N=234 |

*Specification 4 Monthly fixed effects female and per-person income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.068 | 0.257 [0.163] N=2,728 | **0.300* [0.171] N=2,517** | **0.346* [0.188] N=2,249** | **0.455** [0.194] N=1,936** | **0.409* [0.217] N=1,598** |
| WHO weight-for-age z score | 0.307 | -0.086 [0.092] N=3,017 | -0.072 [0.101] N=2,784 | -0.012 [0.102] N=2,490 | -0.048 [0.113] N=2,138 | -0.056 [0.121] N=1,762 |
| WHO weight-for-length/height z score, 0-60 months | 0.351 | 0.049 [0.157] N=2,055 | 0.040 [0.161] N=1,908 | 0.047 [0.171] N=1,720 | -0.142 [0.188] N=1,474 | -0.092 [0.200] N=1,208 |
| WHO BMI-for-age z score | 0.353 | 0.019 [0.145] N=2,642 | 0.010 [0.147] N=2,441 | 0.029 [0.159] N=2,177 | -0.221 [0.168] N=1,871 | -0.206 [0.180] N=1,544 |
| Stunted: HAZ below -2 | 0.187 P=494 | -0.027 [0.032] N=2,728 | -0.030 [0.033] N=2,517 | -0.030 [0.034] N=2,249 | -0.048 [0.036] N=1,936 | -0.033 [0.041] N=1,598 |
| Underweight: WAZ below -2 | 0.054 P=164 | 0.019 [0.016] N=3,017 | 0.016 [0.017] N=2,784 | 0.009 [0.018] N=2,490 | -0.001 [0.020] N=2,138 | -0.006 [0.024] N=1,762 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=230 | -0.021 [0.026] N=2,055 | -0.015 [0.027] N=1,908 | -0.017 [0.029] N=1,720 | 0.002 [0.031] N=1,474 | -0.018 [0.032] N=1,208 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.199 P=403 | 0.018 [0.036] N=2,055 | 0.016 [0.038] N=1,908 | 0.019 [0.040] N=1,720 | -0.006 [0.045] N=1,474 | -0.015 [0.051] N=1,208 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.092 P=191 | -0.005 [0.028] N=2,055 | -0.008 [0.030] N=1,908 | 0.008 [0.032] N=1,720 | -0.008 [0.035] N=1,474 | -0.011 [0.041] N=1,208 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=513 | -0.024 [0.037] N=1,987 | -0.011 [0.038] N=1,845 | -0.006 [0.039] N=1,662 | -0.027 [0.042] N=1,424 | -0.004 [0.045] N=1,167 |
| BMI-for-age z score below -2 | 0.115 P=307 | 0.002 [0.027] N=2,642 | 0.002 [0.028] N=2,441 | 0.002 [0.030] N=2,177 | 0.032 [0.032] N=1,871 | 0.032 [0.035] N=1,544 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.369 P=221 | -0.024 [0.074] N=586 | -0.038 [0.078] N=531 | -0.042 [0.085] N=461 | -0.059 [0.090] N=405 | -0.110 [0.106] N=339 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.222 P=132 | -0.018 [0.062] N=586 | -0.045 [0.064] N=531 | -0.037 [0.073] N=461 | -0.070 [0.075] N=405 | -0.133 [0.083] N=339 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=62 | 0.047 [0.073] N=586 | 0.057 [0.076] N=531 | 0.037 [0.079] N=461 | 0.026 [0.086] N=405 | 0.029 [0.102] N=339 |

*Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.041 [0.213] N=2,289 | -0.030 [0.232] N=2,078 | -0.011 [0.268] N=1,810 | 0.187 [0.288] N=1,497 | -0.188 [0.360] N=1,159 |
| WHO weight-for-age z score | **-0.235* [0.127] N=2,539** | -0.233 [0.147] N=2,306 | -0.141 [0.152] N=2,012 | -0.252 [0.186] N=1,660 | -0.322 [0.228] N=1,284 |
| WHO weight-for-length/height z score, 0-60 months | 0.054 [0.224] N=1,731 | 0.056 [0.239] N=1,584 | 0.032 [0.270] N=1,396 | -0.365 [0.331] N=1,150 | -0.324 [0.416] N=884 |
| WHO BMI-for-age z score | 0.183 [0.213] N=2,216 | 0.216 [0.222] N=2,015 | 0.292 [0.263] N=1,751 | -0.141 [0.302] N=1,445 | -0.062 [0.391] N=1,118 |
| Stunted: HAZ below -2 | 0.035 [0.043] N=2,289 | 0.045 [0.046] N=2,078 | 0.058 [0.049] N=1,810 | 0.037 [0.057] N=1,497 | **0.156** [0.074] N=1,159** |
| Underweight: WAZ below -2 | **0.041* [0.022] N=2,539** | 0.038 [0.025] N=2,306 | 0.027 [0.027] N=2,012 | 0.015 [0.034] N=1,660 | 0.013 [0.045] N=1,284 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.018 [0.037] N=1,731 | -0.012 [0.039] N=1,584 | -0.015 [0.045] N=1,396 | 0.019 [0.056] N=1,150 | -0.025 [0.068] N=884 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.015 [0.044] N=1,731 | 0.012 [0.047] N=1,584 | 0.012 [0.053] N=1,396 | -0.049 [0.065] N=1,150 | -0.091 [0.087] N=884 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.015 [0.031] N=1,731 | -0.019 [0.035] N=1,584 | 0.008 [0.039] N=1,396 | -0.034 [0.045] N=1,150 | -0.066 [0.062] N=884 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.024 [0.050] N=1,672 | 0.046 [0.055] N=1,530 | 0.064 [0.057] N=1,347 | 0.017 [0.068] N=1,109 | 0.121 [0.084] N=852 |
| BMI-for-age z score below -2 | -0.024 [0.032] N=2,216 | -0.036 [0.033] N=2,015 | -0.048 [0.039] N=1,751 | -0.007 [0.043] N=1,445 | -0.016 [0.055] N=1,118 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.059 [0.116] N=481 | 0.060 [0.128] N=426 | 0.092 [0.137] N=356 | 0.103 [0.160] N=300 | 0.051 [0.220] N=234 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.086 [0.092] N=481 | 0.064 [0.097] N=426 | 0.108 [0.112] N=356 | 0.086 [0.125] N=300 | -0.031 [0.145] N=234 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.015 [0.068] N=481 | 0.031 [0.074] N=426 | -0.021 [0.072] N=356 | -0.041 [0.088] N=300 | -0.077 [0.125] N=234 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.076 | **0.273* [0.164] N=2,549** | 0.282 [0.172] N=2,358 | **0.318* [0.186] N=2,101** | **0.392** [0.190] N=1,816** | 0.328 [0.211] N=1,494 |
| WHO weight-for-age z score | 0.303 | -0.037 [0.099] N=2,814 | -0.022 [0.105] N=2,603 | 0.047 [0.107] N=2,322 | -0.003 [0.118] N=2,001 | 0.007 [0.126] N=1,647 |
| WHO weight-for-length/height z score, 0-60 months | 0.362 | 0.143 [0.158] N=1,919 | 0.139 [0.160] N=1,789 | 0.153 [0.169] N=1,610 | -0.027 [0.188] N=1,390 | 0.078 [0.207] N=1,137 |
| WHO BMI-for-age z score | 0.359 | 0.097 [0.146] N=2,470 | 0.090 [0.146] N=2,289 | 0.130 [0.159] N=2,038 | -0.109 [0.166] N=1,760 | -0.072 [0.180] N=1,449 |
| Stunted: HAZ below -2 | 0.187 P=461 | -0.033 [0.033] N=2,549 | -0.033 [0.034] N=2,358 | -0.033 [0.035] N=2,101 | -0.045 [0.036] N=1,816 | -0.022 [0.041] N=1,494 |
| Underweight: WAZ below -2 | 0.054 P=154 | 0.011 [0.017] N=2,814 | 0.006 [0.018] N=2,603 | -0.001 [0.020] N=2,322 | -0.007 [0.022] N=2,001 | -0.013 [0.026] N=1,647 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=214 | -0.034 [0.027] N=1,919 | -0.030 [0.027] N=1,789 | -0.030 [0.029] N=1,610 | -0.011 [0.031] N=1,390 | -0.029 [0.032] N=1,137 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.200 P=378 | 0.023 [0.037] N=1,919 | 0.020 [0.039] N=1,789 | 0.027 [0.041] N=1,610 | -0.000 [0.046] N=1,390 | 0.003 [0.055] N=1,137 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=184 | -0.008 [0.029] N=1,919 | -0.008 [0.032] N=1,789 | 0.011 [0.033] N=1,610 | -0.002 [0.037] N=1,390 | 0.003 [0.043] N=1,137 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=479 | -0.044 [0.040] N=1,855 | -0.033 [0.041] N=1,729 | -0.030 [0.042] N=1,554 | -0.051 [0.045] N=1,342 | -0.028 [0.050] N=1,097 |
| BMI-for-age z score below -2 | 0.113 P=282 | -0.010 [0.026] N=2,470 | -0.010 [0.027] N=2,289 | -0.014 [0.030] N=2,038 | 0.014 [0.032] N=1,760 | 0.009 [0.036] N=1,449 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.360 P=204 | -0.024 [0.081] N=552 | -0.028 [0.084] N=501 | -0.030 [0.092] N=434 | -0.067 [0.096] N=380 | -0.131 [0.115] N=317 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.214 P=121 | -0.028 [0.065] N=552 | -0.039 [0.067] N=501 | -0.024 [0.079] N=434 | -0.073 [0.080] N=380 | **-0.161* [0.091] N=317** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=58 | 0.041 [0.076] N=552 | 0.053 [0.080] N=501 | 0.029 [0.083] N=434 | 0.025 [0.092] N=380 | 0.003 [0.113] N=317 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.077 [0.226] N=2,140 | 0.057 [0.248] N=1,949 | 0.072 [0.279] N=1,692 | 0.197 [0.307] N=1,407 | -0.133 [0.415] N=1,085 |
| WHO weight-for-age z score | -0.216 [0.134] N=2,368 | -0.211 [0.147] N=2,157 | -0.121 [0.157] N=1,876 | -0.282 [0.191] N=1,555 | -0.313 [0.228] N=1,201 |
| WHO weight-for-length/height z score, 0-60 months | 0.049 [0.232] N=1,615 | 0.026 [0.240] N=1,485 | 0.010 [0.270] N=1,306 | -0.355 [0.341] N=1,086 | -0.222 [0.427] N=833 |
| WHO BMI-for-age z score | 0.169 [0.220] N=2,072 | 0.186 [0.225] N=1,891 | 0.291 [0.265] N=1,640 | -0.108 [0.309] N=1,362 | -0.007 [0.402] N=1,051 |
| Stunted: HAZ below -2 | 0.013 [0.045] N=2,140 | 0.022 [0.049] N=1,949 | 0.037 [0.052] N=1,692 | 0.028 [0.062] N=1,407 | **0.149* [0.083] N=1,085** |
| Underweight: WAZ below -2 | 0.038 [0.024] N=2,368 | 0.034 [0.026] N=2,157 | 0.026 [0.029] N=1,876 | 0.025 [0.035] N=1,555 | 0.027 [0.045] N=1,201 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.035 [0.037] N=1,615 | -0.028 [0.038] N=1,485 | -0.028 [0.043] N=1,306 | -0.004 [0.055] N=1,086 | -0.063 [0.059] N=833 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.009 [0.046] N=1,615 | -0.000 [0.049] N=1,485 | 0.007 [0.054] N=1,306 | -0.059 [0.068] N=1,086 | -0.084 [0.094] N=833 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.034 [0.033] N=1,615 | -0.038 [0.037] N=1,485 | -0.007 [0.040] N=1,306 | -0.041 [0.048] N=1,086 | -0.061 [0.068] N=833 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.002 [0.053] N=1,560 | 0.018 [0.057] N=1,434 | 0.041 [0.062] N=1,259 | -0.008 [0.074] N=1,047 | 0.074 [0.096] N=802 |
| BMI-for-age z score below -2 | -0.021 [0.032] N=2,072 | -0.029 [0.032] N=1,891 | -0.043 [0.037] N=1,640 | -0.007 [0.042] N=1,362 | -0.028 [0.051] N=1,051 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.056 [0.129] N=454 | 0.077 [0.137] N=403 | 0.110 [0.150] N=336 | 0.101 [0.172] N=282 | 0.011 [0.239] N=219 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.060 [0.094] N=454 | 0.065 [0.098] N=403 | 0.124 [0.117] N=336 | 0.083 [0.123] N=282 | -0.110 [0.159] N=219 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.028 [0.071] N=454 | 0.036 [0.076] N=403 | -0.020 [0.076] N=336 | -0.050 [0.094] N=282 | -0.138 [0.134] N=219 |

*Specification 6 Monthly fixed effects female and any aid (0/1)  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.076 | **0.276* [0.162] N=2,549** | 0.281 [0.170] N=2,358 | **0.313* [0.185] N=2,101** | **0.392** [0.189] N=1,816** | 0.342 [0.208] N=1,494 |
| WHO weight-for-age z score | 0.303 | -0.026 [0.098] N=2,814 | -0.014 [0.105] N=2,603 | 0.056 [0.106] N=2,322 | 0.008 [0.117] N=2,001 | 0.022 [0.125] N=1,647 |
| WHO weight-for-length/height z score, 0-60 months | 0.362 | 0.138 [0.158] N=1,919 | 0.135 [0.159] N=1,789 | 0.142 [0.168] N=1,610 | -0.037 [0.186] N=1,390 | 0.058 [0.206] N=1,137 |
| WHO BMI-for-age z score | 0.359 | 0.099 [0.145] N=2,470 | 0.091 [0.145] N=2,289 | 0.132 [0.158] N=2,038 | -0.108 [0.166] N=1,760 | -0.077 [0.181] N=1,449 |
| Stunted: HAZ below -2 | 0.187 P=461 | -0.034 [0.032] N=2,549 | -0.034 [0.033] N=2,358 | -0.033 [0.035] N=2,101 | -0.045 [0.036] N=1,816 | -0.025 [0.041] N=1,494 |
| Underweight: WAZ below -2 | 0.054 P=154 | 0.009 [0.017] N=2,814 | 0.004 [0.018] N=2,603 | -0.004 [0.019] N=2,322 | -0.010 [0.022] N=2,001 | -0.017 [0.026] N=1,647 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=214 | -0.034 [0.027] N=1,919 | -0.029 [0.027] N=1,789 | -0.029 [0.029] N=1,610 | -0.011 [0.031] N=1,390 | -0.031 [0.032] N=1,137 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.200 P=378 | 0.022 [0.037] N=1,919 | 0.019 [0.039] N=1,789 | 0.024 [0.041] N=1,610 | -0.004 [0.046] N=1,390 | -0.002 [0.055] N=1,137 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=184 | -0.008 [0.029] N=1,919 | -0.008 [0.031] N=1,789 | 0.010 [0.033] N=1,610 | -0.003 [0.037] N=1,390 | 0.000 [0.043] N=1,137 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.262 P=479 | -0.044 [0.040] N=1,855 | -0.034 [0.041] N=1,729 | -0.028 [0.042] N=1,554 | -0.049 [0.045] N=1,342 | -0.028 [0.049] N=1,097 |
| BMI-for-age z score below -2 | 0.113 P=282 | -0.010 [0.026] N=2,470 | -0.010 [0.027] N=2,289 | -0.014 [0.030] N=2,038 | 0.014 [0.032] N=1,760 | 0.009 [0.036] N=1,449 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.360 P=204 | -0.018 [0.079] N=552 | -0.024 [0.083] N=501 | -0.033 [0.089] N=434 | -0.066 [0.093] N=380 | -0.133 [0.112] N=317 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.214 P=121 | -0.028 [0.064] N=552 | -0.042 [0.065] N=501 | -0.025 [0.077] N=434 | -0.074 [0.078] N=380 | **-0.150* [0.087] N=317** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.105 P=58 | 0.040 [0.076] N=552 | 0.052 [0.081] N=501 | 0.028 [0.084] N=434 | 0.022 [0.093] N=380 | 0.011 [0.114] N=317 |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.076 [0.225] N=2,140 | 0.047 [0.248] N=1,949 | 0.058 [0.279] N=1,692 | 0.188 [0.308] N=1,407 | -0.106 [0.411] N=1,085 |
| WHO weight-for-age z score | -0.202 [0.133] N=2,368 | -0.203 [0.147] N=2,157 | -0.105 [0.154] N=1,876 | -0.258 [0.186] N=1,555 | -0.280 [0.223] N=1,201 |
| WHO weight-for-length/height z score, 0-60 months | 0.072 [0.233] N=1,615 | 0.058 [0.241] N=1,485 | 0.020 [0.270] N=1,306 | -0.348 [0.343] N=1,086 | -0.231 [0.435] N=833 |
| WHO BMI-for-age z score | 0.182 [0.220] N=2,072 | 0.199 [0.224] N=1,891 | 0.300 [0.263] N=1,640 | -0.102 [0.306] N=1,362 | -0.003 [0.400] N=1,051 |
| Stunted: HAZ below -2 | 0.012 [0.045] N=2,140 | 0.023 [0.049] N=1,949 | 0.036 [0.051] N=1,692 | 0.023 [0.060] N=1,407 | **0.138* [0.081] N=1,085** |
| Underweight: WAZ below -2 | 0.036 [0.023] N=2,368 | 0.033 [0.026] N=2,157 | 0.022 [0.028] N=1,876 | 0.020 [0.035] N=1,555 | 0.020 [0.045] N=1,201 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.035 [0.037] N=1,615 | -0.030 [0.039] N=1,485 | -0.029 [0.044] N=1,306 | -0.006 [0.055] N=1,086 | -0.068 [0.061] N=833 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.015 [0.046] N=1,615 | 0.008 [0.049] N=1,485 | 0.010 [0.054] N=1,306 | -0.055 [0.069] N=1,086 | -0.084 [0.096] N=833 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.032 [0.033] N=1,615 | -0.036 [0.037] N=1,485 | -0.005 [0.040] N=1,306 | -0.042 [0.048] N=1,086 | -0.063 [0.068] N=833 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.003 [0.053] N=1,560 | 0.017 [0.058] N=1,434 | 0.039 [0.061] N=1,259 | -0.016 [0.073] N=1,047 | 0.063 [0.091] N=802 |
| BMI-for-age z score below -2 | -0.022 [0.032] N=2,072 | -0.030 [0.032] N=1,891 | -0.044 [0.037] N=1,640 | -0.008 [0.042] N=1,362 | -0.030 [0.052] N=1,051 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.065 [0.126] N=454 | 0.083 [0.135] N=403 | 0.109 [0.144] N=336 | 0.114 [0.164] N=282 | 0.021 [0.235] N=219 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.057 [0.092] N=454 | 0.059 [0.096] N=403 | 0.120 [0.112] N=336 | 0.076 [0.121] N=282 | -0.097 [0.153] N=219 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.022 [0.069] N=454 | 0.032 [0.074] N=403 | -0.025 [0.074] N=336 | -0.055 [0.092] N=282 | -0.145 [0.135] N=219 |

**Mother RDD secondary analysis**

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.018 | -0.222 [0.186] N=2,735 | -0.254 [0.197] N=2,481 | -0.308 [0.204] N=2,191 | **-0.384* [0.218] N=1,869** | **-0.453* [0.243] N=1,515** |
| WHO weight-for-age z score | 0.341 | -0.048 [0.099] N=2,995 | -0.021 [0.106] N=2,730 | -0.098 [0.113] N=2,403 | -0.090 [0.122] N=2,042 | -0.063 [0.136] N=1,650 |
| WHO weight-for-length/height z score, 0-60 months | 0.389 | 0.040 [0.182] N=1,912 | 0.072 [0.187] N=1,733 | 0.059 [0.207] N=1,512 | 0.132 [0.223] N=1,287 | 0.057 [0.266] N=1,037 |
| WHO BMI-for-age z score | 0.382 | 0.124 [0.178] N=2,647 | 0.150 [0.185] N=2,405 | 0.175 [0.202] N=2,124 | 0.248 [0.214] N=1,809 | 0.242 [0.243] N=1,467 |
| Stunted: HAZ below -2 | 0.179 P=490 | 0.011 [0.034] N=2,735 | 0.016 [0.035] N=2,481 | 0.006 [0.038] N=2,191 | 0.023 [0.042] N=1,869 | 0.025 [0.048] N=1,515 |
| Underweight: WAZ below -2 | 0.049 P=150 | 0.001 [0.014] N=2,995 | -0.003 [0.015] N=2,730 | 0.002 [0.016] N=2,403 | 0.008 [0.015] N=2,042 | 0.002 [0.015] N=1,650 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.109 P=208 | 0.025 [0.028] N=1,912 | 0.019 [0.029] N=1,733 | 0.031 [0.032] N=1,512 | -0.003 [0.032] N=1,287 | 0.006 [0.037] N=1,037 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.202 P=387 | 0.035 [0.035] N=1,912 | 0.039 [0.036] N=1,733 | 0.034 [0.038] N=1,512 | 0.045 [0.043] N=1,287 | 0.036 [0.049] N=1,037 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.097 P=192 | 0.007 [0.028] N=1,912 | 0.005 [0.029] N=1,733 | -0.000 [0.031] N=1,512 | 0.007 [0.035] N=1,287 | -0.011 [0.038] N=1,037 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.264 P=487 | 0.049 [0.038] N=1,850 | 0.045 [0.040] N=1,674 | 0.042 [0.042] N=1,460 | 0.014 [0.044] N=1,243 | 0.026 [0.047] N=1,000 |
| BMI-for-age z score below -2 | 0.115 P=304 | 0.027 [0.028] N=2,647 | 0.022 [0.030] N=2,405 | 0.015 [0.033] N=2,124 | 0.001 [0.034] N=1,809 | 0.013 [0.040] N=1,467 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.379 P=279 | 0.010 [0.071] N=739 | -0.017 [0.073] N=673 | -0.041 [0.077] N=613 | -0.037 [0.084] N=524 | -0.025 [0.094] N=427 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.225 P=162 | 0.023 [0.062] N=739 | 0.001 [0.065] N=673 | -0.008 [0.069] N=613 | -0.000 [0.074] N=524 | 0.006 [0.085] N=427 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.104 P=77 | 0.020 [0.049] N=739 | 0.028 [0.052] N=673 | 0.005 [0.052] N=613 | -0.014 [0.056] N=524 | -0.031 [0.064] N=427 |

*Specification 1 Monthly child-age fixed effects 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.028 [0.234] N=2,344 | -0.047 [0.259] N=2,090 | -0.130 [0.278] N=1,800 | -0.297 [0.320] N=1,478 | -0.305 [0.415] N=1,124 |
| WHO weight-for-age z score | -0.145 [0.132] N=2,574 | -0.102 [0.146] N=2,309 | **-0.299* [0.154] N=1,982** | **-0.349** [0.174] N=1,621** | -0.358 [0.221] N=1,229 |
| WHO weight-for-length/height z score, 0-60 months | -0.186 [0.260] N=1,654 | -0.172 [0.273] N=1,475 | -0.269 [0.321] N=1,254 | -0.176 [0.358] N=1,029 | -0.675 [0.429] N=779 |
| WHO BMI-for-age z score | -0.317 [0.198] N=2,272 | -0.341 [0.208] N=2,030 | -0.401 [0.242] N=1,749 | -0.384 [0.262] N=1,434 | **-0.770** [0.320] N=1,092** |
| Stunted: HAZ below -2 | 0.008 [0.043] N=2,344 | 0.012 [0.046] N=2,090 | -0.007 [0.049] N=1,800 | 0.032 [0.058] N=1,478 | 0.010 [0.080] N=1,124 |
| Underweight: WAZ below -2 | -0.003 [0.024] N=2,574 | -0.012 [0.026] N=2,309 | -0.003 [0.029] N=1,982 | 0.002 [0.032] N=1,621 | -0.018 [0.036] N=1,229 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.077** [0.038] N=1,654** | **0.073* [0.041] N=1,475** | **0.109** [0.046] N=1,254** | 0.056 [0.050] N=1,029 | **0.123** [0.054] N=779** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.024 [0.054] N=1,654 | 0.020 [0.057] N=1,475 | 0.010 [0.064] N=1,254 | 0.031 [0.075] N=1,029 | -0.036 [0.094] N=779 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.010 [0.041] N=1,654 | 0.001 [0.043] N=1,475 | -0.010 [0.048] N=1,254 | 0.003 [0.062] N=1,029 | -0.084 [0.080] N=779 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.101* [0.060] N=1,601** | 0.096 [0.065] N=1,425 | 0.108 [0.074] N=1,211 | 0.068 [0.086] N=994 | 0.091 [0.104] N=751 |
| BMI-for-age z score below -2 | **0.079*** [0.026] N=2,272** | **0.079*** [0.029] N=2,030** | **0.076** [0.034] N=1,749** | **0.066** [0.032] N=1,434** | **0.137*** [0.040] N=1,092** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.067 [0.098] N=621 | -0.124 [0.107] N=555 | **-0.210* [0.115] N=495** | **-0.258* [0.132] N=406** | **-0.346** [0.167] N=309** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.075 [0.079] N=621 | -0.132 [0.082] N=555 | **-0.187** [0.088] N=495** | **-0.221** [0.093] N=406** | **-0.302** [0.138] N=309** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.085 [0.055] N=621 | **0.122** [0.059] N=555** | 0.091 [0.063] N=495 | 0.078 [0.069] N=406 | 0.089 [0.089] N=309 |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.018 | -0.223 [0.186] N=2,735 | -0.256 [0.197] N=2,481 | -0.309 [0.205] N=2,191 | **-0.385* [0.218] N=1,869** | **-0.443* [0.242] N=1,515** |
| WHO weight-for-age z score | 0.341 | -0.048 [0.099] N=2,995 | -0.021 [0.106] N=2,730 | -0.098 [0.113] N=2,403 | -0.090 [0.123] N=2,042 | -0.060 [0.137] N=1,650 |
| WHO weight-for-length/height z score, 0-60 months | 0.389 | 0.045 [0.181] N=1,912 | 0.075 [0.187] N=1,733 | 0.061 [0.206] N=1,512 | 0.133 [0.222] N=1,287 | 0.053 [0.263] N=1,037 |
| WHO BMI-for-age z score | 0.382 | 0.124 [0.178] N=2,647 | 0.150 [0.185] N=2,405 | 0.175 [0.202] N=2,124 | 0.249 [0.214] N=1,809 | 0.242 [0.243] N=1,467 |
| Stunted: HAZ below -2 | 0.179 P=490 | 0.011 [0.034] N=2,735 | 0.016 [0.035] N=2,481 | 0.006 [0.038] N=2,191 | 0.023 [0.042] N=1,869 | 0.024 [0.048] N=1,515 |
| Underweight: WAZ below -2 | 0.049 P=150 | 0.001 [0.014] N=2,995 | -0.003 [0.015] N=2,730 | 0.002 [0.016] N=2,403 | 0.008 [0.015] N=2,042 | 0.002 [0.015] N=1,650 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.109 P=208 | 0.023 [0.028] N=1,912 | 0.017 [0.029] N=1,733 | 0.029 [0.031] N=1,512 | -0.003 [0.032] N=1,287 | 0.007 [0.036] N=1,037 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.202 P=387 | 0.034 [0.035] N=1,912 | 0.037 [0.036] N=1,733 | 0.033 [0.039] N=1,512 | 0.044 [0.044] N=1,287 | 0.037 [0.050] N=1,037 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.097 P=192 | 0.006 [0.029] N=1,912 | 0.004 [0.029] N=1,733 | -0.002 [0.032] N=1,512 | 0.006 [0.036] N=1,287 | -0.009 [0.039] N=1,037 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.264 P=487 | 0.047 [0.039] N=1,850 | 0.043 [0.040] N=1,674 | 0.040 [0.043] N=1,460 | 0.013 [0.044] N=1,243 | 0.027 [0.047] N=1,000 |
| BMI-for-age z score below -2 | 0.115 P=304 | 0.027 [0.028] N=2,647 | 0.022 [0.030] N=2,405 | 0.016 [0.033] N=2,124 | 0.002 [0.034] N=1,809 | 0.015 [0.040] N=1,467 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.379 P=279 | 0.020 [0.071] N=739 | -0.008 [0.074] N=673 | -0.036 [0.077] N=613 | -0.034 [0.084] N=524 | -0.016 [0.095] N=427 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.225 P=162 | 0.032 [0.063] N=739 | 0.010 [0.066] N=673 | -0.002 [0.070] N=613 | 0.003 [0.075] N=524 | 0.011 [0.087] N=427 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.104 P=77 | 0.021 [0.049] N=739 | 0.029 [0.052] N=673 | 0.004 [0.052] N=613 | -0.014 [0.056] N=524 | -0.032 [0.063] N=427 |

*Specification 2 Monthly fixed effects and female 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.032 [0.235] N=2,344 | -0.054 [0.259] N=2,090 | -0.139 [0.277] N=1,800 | -0.312 [0.321] N=1,478 | -0.303 [0.414] N=1,124 |
| WHO weight-for-age z score | -0.145 [0.132] N=2,574 | -0.103 [0.145] N=2,309 | **-0.302* [0.154] N=1,982** | **-0.353** [0.175] N=1,621** | -0.354 [0.221] N=1,229 |
| WHO weight-for-length/height z score, 0-60 months | -0.172 [0.262] N=1,654 | -0.164 [0.276] N=1,475 | -0.266 [0.323] N=1,254 | -0.172 [0.361] N=1,029 | -0.658 [0.434] N=779 |
| WHO BMI-for-age z score | -0.317 [0.198] N=2,272 | -0.342 [0.209] N=2,030 | **-0.403* [0.243] N=1,749** | -0.386 [0.263] N=1,434 | **-0.771** [0.320] N=1,092** |
| Stunted: HAZ below -2 | 0.007 [0.043] N=2,344 | 0.012 [0.047] N=2,090 | -0.007 [0.050] N=1,800 | 0.032 [0.058] N=1,478 | 0.009 [0.080] N=1,124 |
| Underweight: WAZ below -2 | -0.003 [0.024] N=2,574 | -0.012 [0.026] N=2,309 | -0.003 [0.029] N=1,982 | 0.002 [0.032] N=1,621 | -0.018 [0.036] N=1,229 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.071* [0.038] N=1,654** | 0.065 [0.041] N=1,475 | **0.102** [0.046] N=1,254** | 0.048 [0.050] N=1,029 | **0.116** [0.054] N=779** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.022 [0.054] N=1,654 | 0.016 [0.057] N=1,475 | 0.004 [0.064] N=1,254 | 0.022 [0.076] N=1,029 | -0.040 [0.094] N=779 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.006 [0.041] N=1,654 | -0.004 [0.043] N=1,475 | -0.017 [0.049] N=1,254 | -0.005 [0.063] N=1,029 | -0.089 [0.081] N=779 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.093 [0.060] N=1,601 | 0.087 [0.065] N=1,425 | 0.097 [0.075] N=1,211 | 0.056 [0.087] N=994 | 0.084 [0.104] N=751 |
| BMI-for-age z score below -2 | **0.078*** [0.026] N=2,272** | **0.077*** [0.029] N=2,030** | **0.075** [0.034] N=1,749** | **0.063* [0.033] N=1,434** | **0.138*** [0.040] N=1,092** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.052 [0.098] N=621 | -0.111 [0.106] N=555 | **-0.201* [0.115] N=495** | **-0.252* [0.132] N=406** | **-0.319* [0.172] N=309** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.058 [0.081] N=621 | -0.115 [0.084] N=555 | **-0.174* [0.090] N=495** | **-0.210** [0.095] N=406** | **-0.282* [0.144] N=309** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.090 [0.057] N=621 | **0.127** [0.061] N=555** | 0.092 [0.063] N=495 | 0.082 [0.070] N=406 | 0.097 [0.092] N=309 |

*Specification 3 Monthly fixed effects female and household income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.018 | -0.224 [0.183] N=2,735 | -0.249 [0.193] N=2,481 | -0.297 [0.202] N=2,191 | **-0.372* [0.211] N=1,869** | **-0.427* [0.234] N=1,515** |
| WHO weight-for-age z score | 0.341 | -0.048 [0.099] N=2,995 | -0.020 [0.106] N=2,730 | -0.092 [0.114] N=2,403 | -0.091 [0.124] N=2,042 | -0.061 [0.139] N=1,650 |
| WHO weight-for-length/height z score, 0-60 months | 0.389 | 0.054 [0.183] N=1,912 | 0.081 [0.189] N=1,733 | 0.073 [0.210] N=1,512 | 0.136 [0.223] N=1,287 | 0.055 [0.262] N=1,037 |
| WHO BMI-for-age z score | 0.382 | 0.120 [0.178] N=2,647 | 0.146 [0.185] N=2,405 | 0.173 [0.201] N=2,124 | 0.239 [0.212] N=1,809 | 0.229 [0.242] N=1,467 |
| Stunted: HAZ below -2 | 0.179 P=490 | 0.011 [0.033] N=2,735 | 0.014 [0.035] N=2,481 | 0.003 [0.037] N=2,191 | 0.020 [0.041] N=1,869 | 0.020 [0.047] N=1,515 |
| Underweight: WAZ below -2 | 0.049 P=150 | 0.002 [0.014] N=2,995 | -0.003 [0.015] N=2,730 | 0.001 [0.016] N=2,403 | 0.007 [0.015] N=2,042 | 0.001 [0.015] N=1,650 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.109 P=208 | 0.023 [0.028] N=1,912 | 0.016 [0.029] N=1,733 | 0.027 [0.032] N=1,512 | -0.004 [0.032] N=1,287 | 0.006 [0.036] N=1,037 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.202 P=387 | 0.037 [0.036] N=1,912 | 0.039 [0.037] N=1,733 | 0.034 [0.039] N=1,512 | 0.044 [0.044] N=1,287 | 0.038 [0.050] N=1,037 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.097 P=192 | 0.008 [0.029] N=1,912 | 0.005 [0.029] N=1,733 | -0.001 [0.032] N=1,512 | 0.006 [0.036] N=1,287 | -0.009 [0.040] N=1,037 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.264 P=487 | 0.051 [0.038] N=1,850 | 0.044 [0.040] N=1,674 | 0.036 [0.043] N=1,460 | 0.010 [0.044] N=1,243 | 0.023 [0.047] N=1,000 |
| BMI-for-age z score below -2 | 0.115 P=304 | 0.027 [0.028] N=2,647 | 0.022 [0.030] N=2,405 | 0.015 [0.033] N=2,124 | 0.001 [0.034] N=1,809 | 0.014 [0.041] N=1,467 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.379 P=279 | 0.016 [0.072] N=739 | -0.010 [0.074] N=673 | -0.036 [0.078] N=613 | -0.038 [0.085] N=524 | -0.010 [0.095] N=427 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.225 P=162 | 0.027 [0.063] N=739 | 0.007 [0.067] N=673 | -0.006 [0.071] N=613 | -0.006 [0.076] N=524 | 0.007 [0.087] N=427 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.104 P=77 | 0.017 [0.048] N=739 | 0.024 [0.051] N=673 | 0.003 [0.051] N=613 | -0.014 [0.056] N=524 | -0.039 [0.062] N=427 |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.024 [0.234] N=2,344 | -0.039 [0.259] N=2,090 | -0.106 [0.278] N=1,800 | -0.273 [0.316] N=1,478 | -0.255 [0.405] N=1,124 |
| WHO weight-for-age z score | -0.144 [0.132] N=2,574 | -0.103 [0.146] N=2,309 | **-0.288* [0.156] N=1,982** | **-0.351** [0.174] N=1,621** | -0.358 [0.221] N=1,229 |
| WHO weight-for-length/height z score, 0-60 months | -0.173 [0.265] N=1,654 | -0.164 [0.280] N=1,475 | -0.266 [0.326] N=1,254 | -0.188 [0.359] N=1,029 | -0.668 [0.430] N=779 |
| WHO BMI-for-age z score | -0.325 [0.198] N=2,272 | -0.348 [0.210] N=2,030 | **-0.409* [0.243] N=1,749** | -0.406 [0.263] N=1,434 | **-0.793** [0.322] N=1,092** |
| Stunted: HAZ below -2 | 0.006 [0.042] N=2,344 | 0.010 [0.046] N=2,090 | -0.013 [0.050] N=1,800 | 0.025 [0.058] N=1,478 | 0.001 [0.079] N=1,124 |
| Underweight: WAZ below -2 | -0.004 [0.023] N=2,574 | -0.012 [0.025] N=2,309 | -0.006 [0.029] N=1,982 | -0.001 [0.032] N=1,621 | -0.021 [0.037] N=1,229 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.072* [0.039] N=1,654** | 0.064 [0.042] N=1,475 | **0.098** [0.048] N=1,254** | 0.045 [0.051] N=1,029 | **0.114** [0.056] N=779** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.025 [0.055] N=1,654 | 0.018 [0.058] N=1,475 | 0.000 [0.065] N=1,254 | 0.016 [0.077] N=1,029 | -0.041 [0.095] N=779 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.007 [0.041] N=1,654 | -0.004 [0.043] N=1,475 | -0.019 [0.049] N=1,254 | -0.008 [0.063] N=1,029 | -0.090 [0.081] N=779 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.096 [0.060] N=1,601 | 0.085 [0.066] N=1,425 | 0.087 [0.076] N=1,211 | 0.045 [0.088] N=994 | 0.079 [0.104] N=751 |
| BMI-for-age z score below -2 | **0.077*** [0.027] N=2,272** | **0.075** [0.029] N=2,030** | **0.073** [0.034] N=1,749** | **0.062* [0.033] N=1,434** | **0.136*** [0.041] N=1,092** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.062 [0.099] N=621 | -0.116 [0.108] N=555 | **-0.204* [0.116] N=495** | **-0.262* [0.137] N=406** | **-0.330* [0.177] N=309** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.064 [0.081] N=621 | -0.118 [0.085] N=555 | **-0.176* [0.090] N=495** | **-0.216** [0.096] N=406** | **-0.289** [0.142] N=309** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.087 [0.056] N=621 | **0.121** [0.060] N=555** | 0.092 [0.063] N=495 | 0.085 [0.069] N=406 | 0.091 [0.090] N=309 |

*Specification 4 Monthly fixed effects female and per-person income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.018 | -0.219 [0.183] N=2,735 | -0.241 [0.194] N=2,481 | -0.289 [0.204] N=2,191 | **-0.358* [0.214] N=1,869** | **-0.414* [0.235] N=1,515** |
| WHO weight-for-age z score | 0.341 | -0.049 [0.097] N=2,995 | -0.018 [0.104] N=2,730 | -0.090 [0.111] N=2,403 | -0.080 [0.120] N=2,042 | -0.059 [0.133] N=1,650 |
| WHO weight-for-length/height z score, 0-60 months | 0.389 | 0.052 [0.182] N=1,912 | 0.081 [0.188] N=1,733 | 0.068 [0.208] N=1,512 | 0.144 [0.223] N=1,287 | 0.067 [0.265] N=1,037 |
| WHO BMI-for-age z score | 0.382 | 0.120 [0.178] N=2,647 | 0.147 [0.185] N=2,405 | 0.171 [0.202] N=2,124 | 0.244 [0.213] N=1,809 | 0.227 [0.243] N=1,467 |
| Stunted: HAZ below -2 | 0.179 P=490 | 0.011 [0.033] N=2,735 | 0.014 [0.035] N=2,481 | 0.003 [0.038] N=2,191 | 0.019 [0.041] N=1,869 | 0.019 [0.047] N=1,515 |
| Underweight: WAZ below -2 | 0.049 P=150 | 0.001 [0.014] N=2,995 | -0.003 [0.015] N=2,730 | 0.001 [0.016] N=2,403 | 0.006 [0.015] N=2,042 | 0.000 [0.015] N=1,650 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.109 P=208 | 0.023 [0.028] N=1,912 | 0.016 [0.029] N=1,733 | 0.029 [0.032] N=1,512 | -0.004 [0.032] N=1,287 | 0.006 [0.036] N=1,037 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.202 P=387 | 0.035 [0.035] N=1,912 | 0.038 [0.037] N=1,733 | 0.034 [0.039] N=1,512 | 0.045 [0.044] N=1,287 | 0.039 [0.050] N=1,037 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.097 P=192 | 0.009 [0.029] N=1,912 | 0.006 [0.029] N=1,733 | 0.001 [0.032] N=1,512 | 0.009 [0.036] N=1,287 | -0.006 [0.040] N=1,037 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.264 P=487 | 0.050 [0.038] N=1,850 | 0.043 [0.039] N=1,674 | 0.036 [0.042] N=1,460 | 0.010 [0.043] N=1,243 | 0.027 [0.046] N=1,000 |
| BMI-for-age z score below -2 | 0.115 P=304 | 0.026 [0.028] N=2,647 | 0.020 [0.030] N=2,405 | 0.014 [0.033] N=2,124 | -0.001 [0.034] N=1,809 | 0.011 [0.040] N=1,467 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.379 P=279 | 0.015 [0.073] N=739 | -0.011 [0.076] N=673 | -0.040 [0.079] N=613 | -0.038 [0.086] N=524 | -0.020 [0.097] N=427 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.225 P=162 | 0.021 [0.063] N=739 | -0.003 [0.067] N=673 | -0.016 [0.070] N=613 | -0.018 [0.075] N=524 | -0.017 [0.085] N=427 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.104 P=77 | 0.014 [0.048] N=739 | 0.024 [0.050] N=673 | 0.000 [0.050] N=613 | -0.019 [0.053] N=524 | -0.050 [0.058] N=427 |

*Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.029 [0.233] N=2,344 | -0.043 [0.260] N=2,090 | -0.098 [0.281] N=1,800 | -0.254 [0.321] N=1,478 | -0.251 [0.407] N=1,124 |
| WHO weight-for-age z score | -0.131 [0.129] N=2,574 | -0.086 [0.142] N=2,309 | **-0.267* [0.152] N=1,982** | **-0.307* [0.171] N=1,621** | -0.313 [0.214] N=1,229 |
| WHO weight-for-length/height z score, 0-60 months | -0.171 [0.263] N=1,654 | -0.167 [0.275] N=1,475 | -0.283 [0.318] N=1,254 | -0.191 [0.354] N=1,029 | -0.658 [0.429] N=779 |
| WHO BMI-for-age z score | -0.318 [0.198] N=2,272 | -0.340 [0.209] N=2,030 | **-0.411* [0.244] N=1,749** | -0.397 [0.264] N=1,434 | **-0.784** [0.324] N=1,092** |
| Stunted: HAZ below -2 | 0.007 [0.042] N=2,344 | 0.010 [0.046] N=2,090 | -0.016 [0.049] N=1,800 | 0.019 [0.058] N=1,478 | -0.002 [0.078] N=1,124 |
| Underweight: WAZ below -2 | -0.006 [0.023] N=2,574 | -0.015 [0.025] N=2,309 | -0.008 [0.030] N=1,982 | -0.005 [0.032] N=1,621 | -0.025 [0.037] N=1,229 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.072* [0.038] N=1,654** | 0.065 [0.041] N=1,475 | **0.105** [0.046] N=1,254** | 0.050 [0.050] N=1,029 | **0.119** [0.054] N=779** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.022 [0.054] N=1,654 | 0.014 [0.058] N=1,475 | -0.003 [0.064] N=1,254 | 0.013 [0.076] N=1,029 | -0.042 [0.095] N=779 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.008 [0.041] N=1,654 | -0.003 [0.043] N=1,475 | -0.018 [0.049] N=1,254 | -0.006 [0.063] N=1,029 | -0.089 [0.082] N=779 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.095 [0.059] N=1,601 | 0.082 [0.065] N=1,425 | 0.082 [0.075] N=1,211 | 0.037 [0.086] N=994 | 0.079 [0.101] N=751 |
| BMI-for-age z score below -2 | **0.078*** [0.027] N=2,272** | **0.076** [0.030] N=2,030** | **0.075** [0.034] N=1,749** | **0.063* [0.033] N=1,434** | **0.139*** [0.040] N=1,092** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.065 [0.099] N=621 | -0.118 [0.109] N=555 | **-0.212* [0.116] N=495** | **-0.266** [0.133] N=406** | **-0.345** [0.172] N=309** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.061 [0.080] N=621 | -0.118 [0.084] N=555 | **-0.178** [0.089] N=495** | **-0.212** [0.093] N=406** | **-0.286** [0.137] N=309** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.084 [0.056] N=621 | **0.123** [0.059] N=555** | 0.092 [0.062] N=495 | 0.083 [0.068] N=406 | 0.077 [0.084] N=309 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.032 | -0.245 [0.189] N=2,565 | -0.260 [0.202] N=2,328 | -0.321 [0.210] N=2,058 | **-0.396* [0.222] N=1,749** | **-0.512** [0.248] N=1,419** |
| WHO weight-for-age z score | 0.342 | -0.022 [0.101] N=2,804 | 0.016 [0.108] N=2,556 | -0.063 [0.116] N=2,253 | -0.059 [0.124] N=1,911 | -0.064 [0.136] N=1,548 |
| WHO weight-for-length/height z score, 0-60 months | 0.394 | 0.104 [0.193] N=1,788 | 0.151 [0.201] N=1,621 | 0.107 [0.221] N=1,415 | 0.173 [0.236] N=1,203 | 0.076 [0.277] N=973 |
| WHO BMI-for-age z score | 0.397 | 0.111 [0.186] N=2,484 | 0.142 [0.195] N=2,258 | 0.139 [0.212] N=1,996 | 0.199 [0.224] N=1,696 | 0.163 [0.258] N=1,378 |
| Stunted: HAZ below -2 | 0.180 P=461 | 0.013 [0.032] N=2,565 | 0.020 [0.034] N=2,328 | 0.011 [0.035] N=2,058 | 0.030 [0.038] N=1,749 | 0.041 [0.043] N=1,419 |
| Underweight: WAZ below -2 | 0.049 P=141 | -0.002 [0.015] N=2,804 | -0.007 [0.015] N=2,556 | -0.005 [0.016] N=2,253 | 0.005 [0.016] N=1,911 | 0.003 [0.016] N=1,548 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=195 | 0.025 [0.029] N=1,788 | 0.016 [0.030] N=1,621 | 0.030 [0.032] N=1,415 | 0.001 [0.033] N=1,203 | 0.013 [0.036] N=973 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.203 P=364 | 0.040 [0.036] N=1,788 | 0.043 [0.038] N=1,621 | 0.034 [0.039] N=1,415 | 0.044 [0.043] N=1,203 | 0.034 [0.049] N=973 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.099 P=184 | 0.008 [0.029] N=1,788 | 0.006 [0.029] N=1,621 | 0.001 [0.031] N=1,415 | 0.013 [0.035] N=1,203 | 0.003 [0.038] N=973 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.265 P=459 | 0.057 [0.040] N=1,730 | 0.052 [0.042] N=1,566 | 0.048 [0.043] N=1,367 | 0.020 [0.045] N=1,161 | 0.041 [0.047] N=937 |
| BMI-for-age z score below -2 | 0.112 P=277 | 0.025 [0.028] N=2,484 | 0.019 [0.029] N=2,258 | 0.014 [0.032] N=1,996 | 0.004 [0.033] N=1,696 | 0.020 [0.038] N=1,378 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.384 P=268 | 0.003 [0.072] N=702 | -0.024 [0.075] N=640 | -0.040 [0.079] N=583 | -0.044 [0.087] N=496 | -0.014 [0.100] N=403 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.231 P=158 | 0.015 [0.066] N=702 | -0.007 [0.070] N=640 | -0.021 [0.074] N=583 | -0.023 [0.079] N=496 | -0.007 [0.094] N=403 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.099 P=69 | 0.025 [0.050] N=702 | 0.035 [0.052] N=640 | 0.011 [0.052] N=583 | -0.009 [0.057] N=496 | -0.022 [0.063] N=403 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| **Outcome** | **8 years** | **7 years** | **6 years** | **5 years** | **4 years** |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.023 [0.231] N=2,199 | 0.035 [0.256] N=1,962 | -0.056 [0.274] N=1,692 | -0.184 [0.313] N=1,383 | -0.227 [0.389] N=1,053 |
| WHO weight-for-age z score | -0.130 [0.135] N=2,410 | -0.075 [0.148] N=2,162 | **-0.282* [0.156] N=1,859** | **-0.326* [0.174] N=1,517** | **-0.374* [0.213] N=1,154** |
| WHO weight-for-length/height z score, 0-60 months | -0.134 [0.277] N=1,545 | -0.115 [0.293] N=1,378 | -0.277 [0.343] N=1,172 | -0.205 [0.374] N=960 | -0.703 [0.455] N=730 |
| WHO BMI-for-age z score | -0.322 [0.201] N=2,130 | -0.333 [0.213] N=1,904 | **-0.427* [0.246] N=1,642** | -0.416 [0.257] N=1,342 | **-0.834*** [0.305] N=1,024** |
| Stunted: HAZ below -2 | -0.008 [0.044] N=2,199 | -0.003 [0.048] N=1,962 | -0.021 [0.051] N=1,692 | 0.010 [0.058] N=1,383 | -0.002 [0.077] N=1,053 |
| Underweight: WAZ below -2 | -0.008 [0.025] N=2,410 | -0.017 [0.027] N=2,162 | -0.014 [0.031] N=1,859 | -0.004 [0.034] N=1,517 | -0.022 [0.040] N=1,154 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.074* [0.040] N=1,545** | 0.065 [0.043] N=1,378 | **0.102** [0.048] N=1,172** | 0.050 [0.054] N=960 | **0.109* [0.058] N=730** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.022 [0.056] N=1,545 | 0.014 [0.060] N=1,378 | -0.007 [0.066] N=1,172 | 0.001 [0.077] N=960 | -0.058 [0.097] N=730 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.006 [0.043] N=1,545 | -0.018 [0.044] N=1,378 | -0.033 [0.051] N=1,172 | -0.019 [0.065] N=960 | -0.092 [0.085] N=730 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.093 [0.065] N=1,496 | 0.084 [0.071] N=1,332 | 0.091 [0.079] N=1,133 | 0.026 [0.096] N=927 | 0.047 [0.118] N=703 |
| BMI-for-age z score below -2 | **0.077*** [0.028] N=2,130** | **0.074** [0.030] N=1,904** | **0.072** [0.035] N=1,642** | **0.062* [0.035] N=1,342** | **0.139*** [0.040] N=1,024** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.079 [0.097] N=590 | -0.138 [0.104] N=528 | **-0.206* [0.112] N=471** | **-0.256* [0.130] N=384** | **-0.307* [0.171] N=291** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.080 [0.083] N=590 | -0.135 [0.087] N=528 | **-0.195** [0.092] N=471** | **-0.229** [0.099] N=384** | **-0.312** [0.146] N=291** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.091 [0.057] N=590 | **0.129** [0.061] N=528** | 0.091 [0.062] N=471 | 0.072 [0.069] N=384 | 0.082 [0.093] N=291 |

*Specification 6 Monthly fixed effects female and any aid (0/1)  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.032 | -0.252 [0.189] N=2,565 | -0.266 [0.202] N=2,328 | -0.326 [0.209] N=2,058 | **-0.398* [0.223] N=1,749** | **-0.504** [0.248] N=1,419** |
| WHO weight-for-age z score | 0.342 | -0.028 [0.101] N=2,804 | 0.009 [0.108] N=2,556 | -0.075 [0.116] N=2,253 | -0.066 [0.125] N=1,911 | -0.062 [0.138] N=1,548 |
| WHO weight-for-length/height z score, 0-60 months | 0.394 | 0.106 [0.193] N=1,788 | 0.149 [0.201] N=1,621 | 0.108 [0.221] N=1,415 | 0.179 [0.235] N=1,203 | 0.080 [0.280] N=973 |
| WHO BMI-for-age z score | 0.397 | 0.113 [0.186] N=2,484 | 0.145 [0.194] N=2,258 | 0.139 [0.211] N=1,996 | 0.199 [0.224] N=1,696 | 0.161 [0.258] N=1,378 |
| Stunted: HAZ below -2 | 0.180 P=461 | 0.013 [0.032] N=2,565 | 0.019 [0.034] N=2,328 | 0.011 [0.035] N=2,058 | 0.029 [0.038] N=1,749 | 0.039 [0.043] N=1,419 |
| Underweight: WAZ below -2 | 0.049 P=141 | -0.001 [0.015] N=2,804 | -0.005 [0.015] N=2,556 | -0.002 [0.016] N=2,253 | 0.006 [0.016] N=1,911 | 0.003 [0.016] N=1,548 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.110 P=195 | 0.023 [0.029] N=1,788 | 0.015 [0.030] N=1,621 | 0.028 [0.032] N=1,415 | -0.004 [0.032] N=1,203 | 0.008 [0.036] N=973 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.203 P=364 | 0.038 [0.036] N=1,788 | 0.041 [0.037] N=1,621 | 0.031 [0.039] N=1,415 | 0.041 [0.043] N=1,203 | 0.031 [0.049] N=973 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.099 P=184 | 0.007 [0.028] N=1,788 | 0.005 [0.028] N=1,621 | -0.001 [0.031] N=1,415 | 0.010 [0.034] N=1,203 | -0.000 [0.037] N=973 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.265 P=459 | 0.053 [0.040] N=1,730 | 0.047 [0.041] N=1,566 | 0.043 [0.043] N=1,367 | 0.012 [0.044] N=1,161 | 0.035 [0.046] N=937 |
| BMI-for-age z score below -2 | 0.112 P=277 | 0.024 [0.028] N=2,484 | 0.018 [0.029] N=2,258 | 0.013 [0.031] N=1,996 | 0.003 [0.032] N=1,696 | 0.018 [0.038] N=1,378 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.384 P=268 | 0.000 [0.072] N=702 | -0.027 [0.074] N=640 | -0.048 [0.078] N=583 | -0.049 [0.086] N=496 | -0.018 [0.098] N=403 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.231 P=158 | 0.016 [0.066] N=702 | -0.007 [0.069] N=640 | -0.021 [0.073] N=583 | -0.024 [0.078] N=496 | -0.004 [0.092] N=403 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.099 P=69 | 0.021 [0.049] N=702 | 0.033 [0.051] N=640 | 0.005 [0.051] N=583 | -0.015 [0.055] N=496 | -0.033 [0.061] N=403 |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.002 [0.228] N=2,199 | 0.015 [0.251] N=1,962 | -0.076 [0.268] N=1,692 | -0.231 [0.309] N=1,383 | -0.263 [0.390] N=1,053 |
| WHO weight-for-age z score | -0.135 [0.134] N=2,410 | -0.080 [0.149] N=2,162 | **-0.296* [0.156] N=1,859** | **-0.332* [0.173] N=1,517** | **-0.379* [0.215] N=1,154** |
| WHO weight-for-length/height z score, 0-60 months | -0.133 [0.274] N=1,545 | -0.113 [0.289] N=1,378 | -0.274 [0.339] N=1,172 | -0.171 [0.374] N=960 | -0.696 [0.446] N=730 |
| WHO BMI-for-age z score | -0.313 [0.201] N=2,130 | -0.326 [0.211] N=1,904 | **-0.425* [0.243] N=1,642** | -0.391 [0.256] N=1,342 | **-0.822*** [0.309] N=1,024** |
| Stunted: HAZ below -2 | -0.005 [0.044] N=2,199 | 0.000 [0.048] N=1,962 | -0.017 [0.050] N=1,692 | 0.018 [0.058] N=1,383 | 0.008 [0.076] N=1,053 |
| Underweight: WAZ below -2 | -0.007 [0.025] N=2,410 | -0.016 [0.027] N=2,162 | -0.011 [0.031] N=1,859 | -0.002 [0.034] N=1,517 | -0.018 [0.040] N=1,154 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.074* [0.039] N=1,545** | 0.065 [0.042] N=1,378 | **0.103** [0.047] N=1,172** | 0.046 [0.052] N=960 | **0.111* [0.056] N=730** |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.023 [0.055] N=1,545 | 0.015 [0.059] N=1,378 | -0.007 [0.065] N=1,172 | 0.005 [0.077] N=960 | -0.059 [0.095] N=730 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.005 [0.042] N=1,545 | -0.017 [0.044] N=1,378 | -0.032 [0.050] N=1,172 | -0.018 [0.065] N=960 | -0.094 [0.083] N=730 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.094 [0.064] N=1,496 | 0.087 [0.071] N=1,332 | 0.097 [0.079] N=1,133 | 0.036 [0.095] N=927 | 0.073 [0.116] N=703 |
| BMI-for-age z score below -2 | **0.076*** [0.027] N=2,130** | **0.074** [0.030] N=1,904** | **0.071** [0.034] N=1,642** | **0.059* [0.034] N=1,342** | **0.138*** [0.040] N=1,024** |
| WHO 2007 BMI overweight above +1, 61+ months | -0.076 [0.097] N=590 | -0.135 [0.104] N=528 | **-0.208* [0.112] N=471** | **-0.252* [0.130] N=384** | **-0.293* [0.173] N=291** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.073 [0.083] N=590 | -0.132 [0.087] N=528 | **-0.188** [0.092] N=471** | **-0.227** [0.098] N=384** | **-0.300** [0.148] N=291** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.088 [0.057] N=590 | **0.128** [0.060] N=528** | 0.081 [0.062] N=471 | 0.063 [0.067] N=384 | 0.060 [0.091] N=291 |

*Notes:* Cells report the OLS policy coefficient, birth-month-clustered standard error in brackets, and N. Exact child-month fixed effects for analysis months 0 to 83 are included but not displayed. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add the categorical number of aid types (0 as reference) and any aid. Aid and income controls are not combined. Aid specifications exclude households with a blank or non-yes/no answer on any of the six main aid questions. Columns use symmetric 8- to 4-year bandwidths around the January 1987 cutoff. Every regression includes reform eligibility and split linear trends. Donut estimates exclude running-variable months -12 through +11 and show only the policy estimate, standard error, and N. Regressions use FAKTOR_FERT child survey weights. The sample retains one latest-born eligible child per household. Significant policy coefficients are bold. P in the mean column is the unweighted number of positive outcomes in the no-donut 8-year sample. *** p<0.01, ** p<0.05, * p<0.10.